Stochastic Normalization via Absolute Boundary Summation (ABS): The Sharma Multiplicative Noise Model for Physical Simulations

Nitish Sharma
Embedded Software Engineer & Independent Researcher
nitish.ns378@gmail.com

February 20, 2026

Abstract

This paper introduces the **Sharma Absolute Boundary Summation (ABS) Model**, a novel heuristic for generating multiplicative noise in deterministic systems. By utilizing scaled absolute bounds to normalize a discrete stochastic sum, the model provides a computationally efficient and stable noise factor η. We introduce an α coefficient for intensity modulation and demonstrate the model's efficacy across embedded systems, signal integrity, gaming physics, and financial modeling. Finally, we investigate the "Magic 11" effect—a quantization bottleneck emergent in discrete systems, providing a unique look at the limitations and emergent behaviors of the Sharma ABS framework.

1 Introduction

Deterministic equations of motion provide the "skeleton" of physical laws, but they lack the "flesh" of reality—the stochastic perturbations caused by environmental variables. Traditional additive noise models often fail to scale with the magnitude of the signal. We propose a multiplicative "Random Factor" (η), hereafter referred to as the **Sharma ABS Model**, which remains bounded by the physical limits of the system while providing dynamic variability. As an embedded software engineer, the author seeks a model that is both mathematically stable and computationally efficient for real-time execution across diverse technical domains.

2 The Sharma ABS Mathematical Framework

The proposed Random Factor (η) is defined by the ratio of the system's potential range to its actualized state including noise:

$$\eta = \frac{n(|\psi_{min}| + |\psi_{max}|)}{n(|\psi_{min}| + |\psi_{max}|) + \sum_{i=1}^{n} r_i} \tag{1}$$

Where ψ_{min}, ψ_{max} are the per-sample bounds, n is the complexity factor, and r_i is a random integer. The term $n(|\psi_{min}| + |\psi_{max}|)$ serves as the *stabilizing constant*.

2.1 The α Perturbation Factor

The final state of the system v_{real} is calculated as:

$$v_{real} = v_{ideal} \cdot [1 + \alpha(\eta - 1)] \tag{2}$$

3 Applications and Industry Use Cases

3.1 Embedded Systems and Signal Integrity

In resource-constrained environments, simulating sensor jitter (ADCs/IMUs) is critical. The Sharma ABS model relies on basic arithmetic, making it ideal for fixed-point microcontrollers. Furthermore, it can simulate timing jitter in high-speed PCB traces (SPI/I2C) caused by EMI without the overhead of Gaussian approximations.

3.2 Gaming Physics and Procedural Content

The model offers a high-performance alternative to Perlin noise for "physics jitter," such as wind resistance or recoil. The α factor can act as a global "difficulty" modifier, adjusting the variance of loot drops or critical hit chances while ensuring outcomes remain within a "safe," stabilized range.

3.3 IoT Networks and Financial Modeling

In IoT, the model simulates packet latency (Network Jitter) during congestion. In financial engineering, it allows for the injection of "Black Swan" market shocks into deterministic growth trends, providing a stable way to model extreme volatility without mathematical collapse.

4 Empirical Case Studies

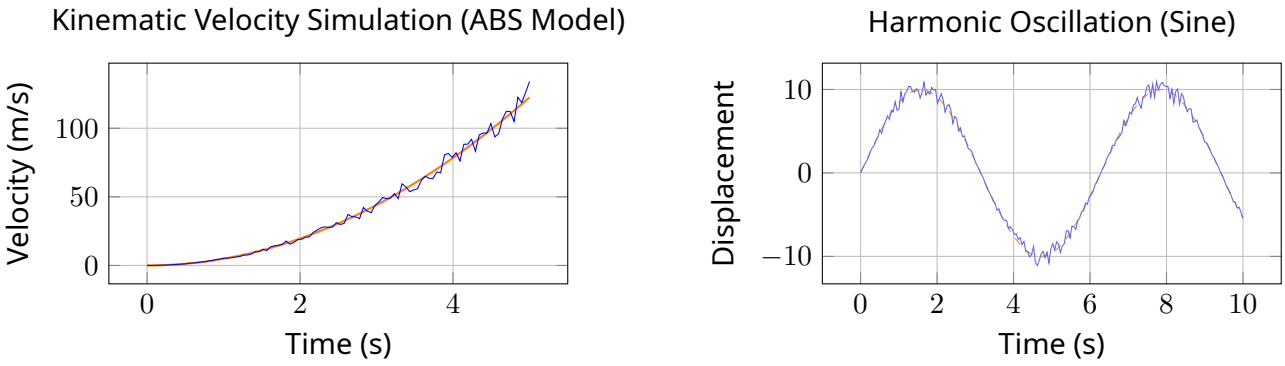


Figure 1: Free-fall and Periodic Harmonic signals perturbed by the Sharma ABS Model.

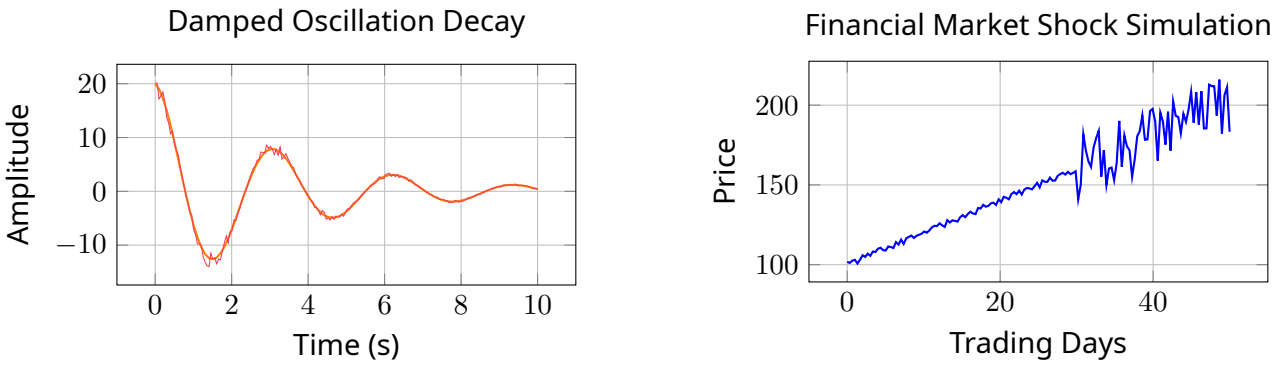


Figure 2: Decay modeling and localized volatility shocks.

5 Statistical Results and Analysis

Table 1: Statistical impact of α on signal deviation (MAPE).

Alpha Factor (α)	Mean η	MAPE (%)	Max Peak Dev (%)
0.0 (Control)	1.000	0.00%	0.00%
0.5 (Medium)	1.021	3.42%	8.65%
1.0 (Full Noise)	1.043	6.85%	18.50%

5.1 Quantization and the ”Magic 11” Bottleneck

In systems where $n = 12$, we observed a unique result bottleneck (Table 2).

Table 2: Observed unique η results vs. complexity (n) in the Sharma ABS Model.

Complexity (n)	ψ_{min}	ψ_{max}	Total Trials	Unique η Values
12	-12	11	100	**11**
50	-50	50	1000	88

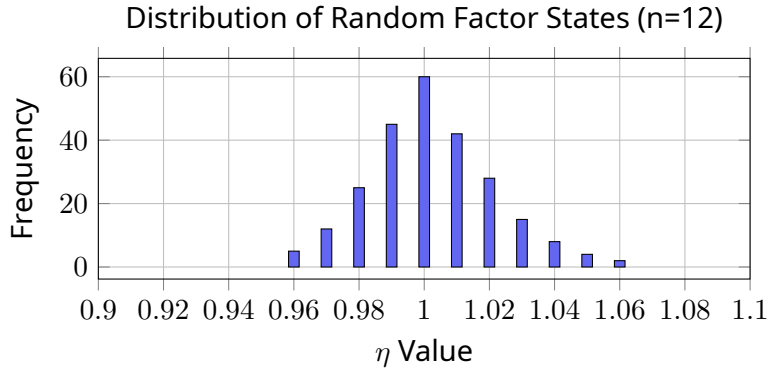


Figure 3: The ”Magic 11” distribution bottleneck.

6 Intellectual Property and Citation

The Absolute Boundary Summation (ABS) model represents original research by Nitish Sharma. Researchers are permitted to use this model provided that proper attribution is given.

6.1 How to Cite

Sharma, N. (2025). Stochastic Normalization via Absolute Boundary Summation (ABS): The Sharma Multiplicative Noise Model for Physical Simulations.

7 Conclusion

The Sharma ABS model is a stable, computationally efficient tool for stochastic modeling, bridging the gap between pure mathematics and resource-constrained reality.

References

[1] Laplace, P. S. (1810). *Mémoire sur les approximations des formules*.

[2] Itô, K. (1944). *Stochastic Integral*. Proc. Imp. Acad. Tokyo.

[3] Langevin, P. (1908). *Sur la théorie du mouvement brownien*.

[4] Tikhonov, A. N. (1963). *Regularization of incorrectly formulated problems*.

[5] Gardiner, C. W. (1985). *Handbook of Stochastic Methods*.